

React Programs

Problem statement 1-Create a functional component that accepts props and displays a personalized message.

Source code:

[App.js](#)

```
import React from "react";
import Greeting from "../Greeting";
function App() {
  return (
    <div>
      <Greeting name="Akshit" />
    </div>
  );
}
export default App;
```

[Greeting.js](#)

```
import React from "react";
function Greeting(props) {
  return (
    <div>
      <h2>Hello, {props.name}!</h2>
      <p>Welcome to React programming </p>
    </div>
  );
}
export default Greeting;
```

OUTPUT:



Problem statement 2-Create a class component that maintains a counter and provides buttons to increase and decrease the counter.

Source code:

[App.js](#)

```
import React from "react"; import
Counter from "../Counter"; function
App() {
  return (
    <div>
      <Counter />
    </div>
  );
}
export default App;
```

[Counter.js](#)

```
import React, { Component } from "react"; class
Counter extends Component { constructor(props) {
  super(props);
  this.state = {
    count: 0,
  };
}
  increase = () => {
```

```

    this.setState({ count: this.state.count + 1 });
  };

  decrease = () => {
    this.setState({ count: this.state.count - 1 });
  };

  render() { return (
    <div style={{ textAlign: "center", marginTop: "50px" }}>
      <h1>Counter App</h1>
      <h2>{this.state.count}</h2>

      <button onClick={this.increase}>Increase +</button>
      <button onClick={this.decrease}>Decrease -</button>

    </div>

  );
}
}

export default Counter;

```

OUTPUT:

Counter App

3

Increase +

Decrease -



Counter App

-4

Increase +

Decrease -

Problem statement 3-Create a functional component that maintains a counter and provides buttons to increase, decrease and reset the counter.

Source code:

[App.js](#)

```
import React, { useState } from "react"; function
App() {
  const [count, setCount] = useState(0); return (
    <div style={{ textAlign: "center", marginTop: "50px" }}>
      <h1>Counter App</h1>
      <h2>{count}</h2>
      <button onClick={() => setCount(count + 1)}>Increase</button>
      <button onClick={() => setCount(count - 1)}>Decrease</button>
      <button onClick={() => setCount(0)}>Reset</button>
    </div>
  );
}
export default App;
```

[Counter.js](#)

```
import React, { useState } from "react"; function
Counter() {
  const [count, setCount] = useState(0); return (
    <div style={{ textAlign: "center", marginTop: "50px" }}>
      <h1>Counter</h1>
```

```
<h2>{count}</h2>

<button onClick={() => setCount(count + 1)}>

  Increase

</button>

<button onClick={() => setCount(count - 1)}>

  Decrease

</button>

<button onClick={() => setCount(0)}> Reset

</button>

</div>

);
}

export default Counter;
```

OUTPUT:

Counter App

6

Increase Decrease Reset

Problem statement 4- Create a functional component that displays a running clock and the current date.

Source Code:

```
import React from "react"; import
Clock from "./Clock"; function
App() {
return (
    <div style={{ textAlign: "center" }}>
    <h1>Clock Application</h1>
    <Clock />
    </div>
);
}
Clock.js
import React, { useState, useEffect } from "react";
function App() {
    const [time, setTime] = useState(new Date());
    useEffect(() => {
        setInterval(() => { setTime(new
            Date());
        }, 1000);
    }, []);
    return (
        <div style={{ textAlign: "center", marginTop: "50px" }}>
        <h1>Clock App</h1>
```

```
<h2>{time.toLocaleTimeString()}</h2>  
  
<p>{time.toDateString()}</p>  
  
</div>  
  
);  
  
}export default App;
```

OUTPUT:



Clock Application

Clock App

6:43:45 PM

Mon Apr 27 2026

Problem statement 5- Create a component with a form that updates the state based on user input.

Source code:

```
import React, { useState } from "react"; function
UserForm() {
  const [formData, setFormData] = useState({ name: "",
    email: ""
  });
  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData({
      ...formData,
      [name]: value
    });
  };
  const handleSubmit = (e) => {
    e.preventDefault();
    alert(`Name: ${formData.name}, Email: ${formData.email}`);
  }; return (
    <div style={{ textAlign: "center", marginTop: "50px" }}>
      <h2>User Form</h2>
      <form onSubmit={handleSubmit}>
        <input
```

```

    type="text" name="name"
    placeholder="Enter Name"
    value={formData.name}
    onChange={handleChange}
  /><br /><br /><input
    type="email" name="email"
    placeholder="Enter Email"
    value={formData.email}
    onChange={handleChange}
  />
  <br /><br /><button type="submit">Submit</button>
</form>
<h3>Live Data:</h3>
<p>Name: {formData.name}</p>
<p>Email: {formData.email}</p>
</div>
);
} ;export default UserForm;

```

OUTPUT:

localhost:3000 says

Name: Apratim Rana, Email: apratimrana234@gmail.com

OK

apratimrana234@gmail.com

Submit

Live Data:

Name: Apratim Rana

Email: apratimrana234@gmail.com

User Form

Apratim Rana

apratimrana234@gmail.com

Submit

Live Data:

Problem statement 6- Create a React application with multiple routes.

Source code:

```
import React from "react";

import { BrowserRouter, Routes, Route, Link } from "react-router-dom"; function

App() {

  function Home() {

    return <h2>Home Page</h2>;

  }

  function About() {

    return <h2>About Page</h2>;

  }

  function Contact() {

    return <h2>Contact Page</h2>;

  }

  return (

    <BrowserRouter>

    <div style={{ textAlign: "center" }}>

      <h1>React App</h1>

      {}

      <nav>

        <Link to="/">Home</Link> |{" "}

        <Link to="/about">About</Link> |{" "}

        <Link to="/contact">Contact</Link>

      </nav>

    </div>

  );

}
```

```
<hr />

{}

<Routes>

  <Route path="/" element={<Home />} />

  <Route path="/about" element={<About />} />

  <Route path="/contact" element={<Contact />} />

</Routes>

</div>

</BrowserRouter>

);

}

export default App;
```

OUT

React App

[Home](#) | [About](#) | [Contact](#)

Contact Page

